Tidal Reach (Island / Ocean / Naval Warfare & Amphibious Operations)

Executive Summary

Tidal Reach is a **tropical island–ocean hybrid battlefield** built for **naval warfare, amphibious landings, littoral combat, and autonomous multi-domain coordination**. The terrain features coral shallows, deep-water channels, sandy beaches, mangrove swamps, cliffside fortifications, island villages, and offshore naval platforms. The map emphasizes Obsidian Protocol's autonomous amphibious routing, naval-air coordination, sensor-fusion across water and land, and logistics under dynamic tide conditions. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10×10 m physical area representing ~1 km² of island-ocean terrain. Key features include landing beaches, naval chokepoints, airspace corridors, reef hazards, and multi-domain combat across sea, air, and land.

Design Goals & Gameplay

Naval Warfare in Littoral Waters

Tidal Reach prioritizes ship-to-ship combat, torpedo lanes, reef navigation, and hull-hazard maneuvering. Coral shelves create dynamic cover and movement obstacles.

Amphibious Operations & Beach Landings

Players conduct coordinated landings using amphibious armor, drones, and infantry. Beaches, mangroves, and cliffs create layered defensive zones.

Air Operations & Reconnaissance

VTOL aircraft, drones, and long-range recon planes operate above the island chain. Airspace corridors between cliffs and offshore platforms define movement and targeting.

Autonomy & Multi-Domain Coordination

Autonomous units coordinate naval, air, and land forces simultaneously. Units adapt to shifting tides, fog pockets, and sonar interference.

Sensor & Information Warfare

Sea spray, humidity, thermal distortion, sonar scatter, and magnetic interference degrade sensors. Relay towers on cliffs restore clarity. Destroying relays creates blind pockets across the island chain.

Logistics & Resources

Fuel caches, island depots, and offshore platforms serve as resupply nodes. Convoys must navigate beach approaches and avoid ambush-prone mangrove corridors.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between naval dominance, air superiority, or amphibious mobility. Blue deploys from the southwest offshore staging platform; Red deploys from the northeast cliffside fortress.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area ≈ 1 km² island-ocean zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit ≈ 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Offshore Staging Platform) — (0,0), 150×150 m Naval docks, drone pads, amphibious armor staging.
Red HQ (Cliffside Fortress Command) — (1000,1000), 150×150 m Hardened cliff bunker with radar mast, AA emplacements, and sonar arrays.
Central Tidal Channel — (500,450), 350×250 m Deep-water lane, coral shelves, naval combat zone.
Island Depot & Village — (250,800), 120×120 m Supply huts, docks, interior flanking routes.
Mangrove Swamp Network — (800,300), 200×200 m Dense vegetation, shallow water, stealth movement.
Landing Beach Corridor — (0,450 → 400,900) Primary amphibious chokepoint; layered defenses and tide-driven hazards.
Cliffside Trail Route — (300,0 → 450,350) Safer land route with partial cover.
Cliff Relay A — (650,150), +240 m Long-range sensor vantage point.
Cliff Relay B — (150,650), +260 m Comms relay hub and sonar control node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Landing Beach Corridor

Major amphibious chokepoint; controlling it determines land access.

Tidal Channel

Long-range naval and air engagements; dynamic coral cover.

Mangrove Swamp Network

Interior flanking routes; stealthy but slow.

Cliffside Trail

Predictable but safe; vulnerable to overwatch from cliffs.

Sightlines

Cliff Relay A provides long-range sensor coverage (~350 m). Sea spray and humidity create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW offshore platform + naval staging.
Red Deployment: NE cliffside fortress + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Beaches, cliffs, coral shelves, mangroves, offshore platforms.

Vegetation

Palm trees, mangroves, shrubs, beach grass.

Buildings & Props

Military: AA guns, radar dishes, sonar arrays, bunkers. Industrial: Docks, generators, pipelines. Logistics: Ships, cargo crates, fuel drums, drones.

Effects

Sea spray, fog, humidity haze, wave impacts, thermal distortion.

Audio

Waves, wind, ship engines, sonar pings, radio chatter.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
Coral/Ice	3	200k	600k
Naval/Inc	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out strait, beaches, mangroves, AR anchor test.
2. **Core Systems — Month 2** NavMesh (sea + land), amphibious autonomy, air-sea coordination.
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity.
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio.
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (sea spray, fog, thermal distortion).
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify naval routing, amphibious landings, air-sea coordination.

Comms/Sensors

Test relay destruction, fog-of-war updates, sonar interference.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, amphibious logic, deployment budget enforcement, anchor drift, sensor accuracy.